

Mini project report

Emotion detection using python

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Introduction

emotion Detection is an Artificial Intelligence application that identifies human emotions from facial expressions using a webcam. The system captures a face, analyses facial features using computer vision and deep learning, and predicts emotions such as Happy, Sad, Angry, Fear, Surprise, Neutral, and Disgust.

Working Principle

1. The webcam continuously captures live video.
2. Each frame is analysed to locate the face.
3. The detected face is processed and passed to the DeepFace model.
4. The model extracts facial features and predicts the most likely emotion.
5. The predicted emotion is displayed above the face in real time.
6. The process repeats until the user exits the application.

Objective

- Detect facial emotions in real time.
- Use a webcam for face detection.
- Display the detected emotion on the screen.
- Learn AI, Machine Learning, and Computer Vision concepts.

Software Requirements

- Python 3.12
- Visual Studio Code
- OpenCV
- TensorFlow
- DeepFace
- NumPy

Hardware Requirements

- Laptop/Desktop
- Webcam
- Minimum 4 GB RAM
- Windows 10/11

Technologies Used

- Python
- OpenCV
- TensorFlow/Keras
- DeepFace
- Computer Vision
- Artificial Intelligence

Add the Methodology

1. Start the program.
2. Open the webcam.
3. Capture video frames.
4. Detect the face.
5. Analyse facial features.
6. Predict the emotion.
7. Display the emotion on the screen.
8. Continue until the user presses **Q** to quit.

Advantages

- Real-time emotion detection
- Easy to use
- Fast processing
- Helps understand user emotions
- Useful in AI applications

Applications

- Online learning
- Healthcare
- Security systems
- Human-computer interaction
- Customer service
- Smart attendance systems

Flowchart

START

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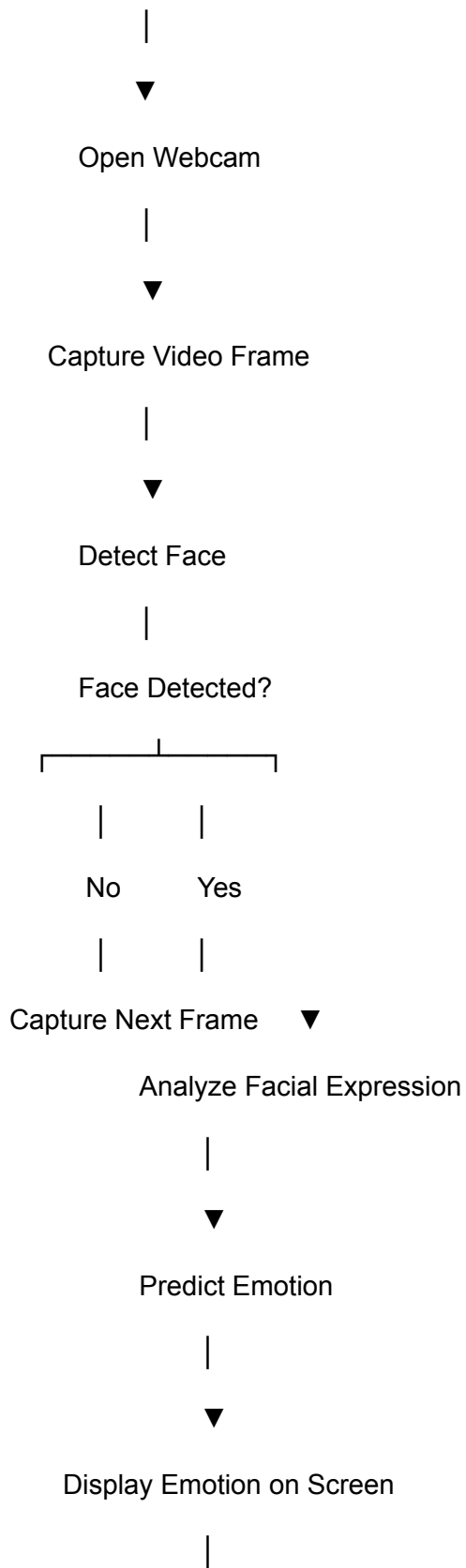
Initialize Python Program

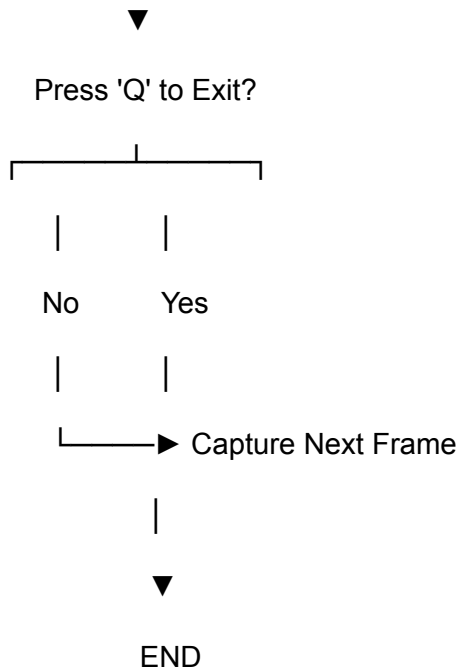
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Import Required Libraries

(OpenCV, DeepFace, TensorFlow, NumPy)





Algorithm

Step 1: Start the program.

Step 2: Import the required Python libraries:

- OpenCV
- DeepFace
- TensorFlow
- NumPy

Step 3: Initialize the webcam using OpenCV.

Step 4: Capture a video frame from the webcam.

Step 5: Detect the face in the captured frame.

Step 6: If no face is detected, capture the next frame.

step7: If a face is detected, preprocess the image for emotion analysis

Step 8: Pass the face image to the DeepFace emotion detection model.

Step 9: Predict the facial emotion (Happy, Sad, Angry, Fear, Surprise, Neutral, etc.).

Step 10: Display the detected emotion on the video frame.

Step 11: Continue processing frames until the user presses the **Q** key.

Step 12: Release the webcam and close all OpenCV windows.

Step 13: End the program.

Conclusion

The Emotion Detection system successfully identifies human emotions from facial expressions using Artificial Intelligence, OpenCV, and DeepFace. It demonstrates how computer vision and deep learning can analyse emotions in real time. This project provides practical experience with AI technologies and has applications in healthcare, education, security, and customer service.